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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes '(use Derived Categories, Proposition ...)' inside the counted proof span, reproduced verbatim. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- The distinguished triangle relating a module, its localization at f, and the kernel of the localization map. used at proof lines 30020-30021. Introduces the object F whose vanishing must be established.
- f-power torsion of the cohomology of the localization kernel. used at proof line 30023. Is the structural fact that makes the kernel of multiplication by f detectable.
- The identification of the cone of f : M -> M with the cone of f : F -> F when f acts invertibly on M_f. invoked at proof lines 30024-30026. Is the step that transfers the hypothesis on C to information about F.

Source reference. Original lemma 05R0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7394–7394. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let R be a ring and $f \in R$. Let $M \in D(R)$ and let C be the cone of $f : M \to M$. If $H^i(M)_f = 0$ for $i < 0$ and $H^i(C) = 0$ for $i < -1$, then $H^i(M) = 0$ for $i < 0$.*

Proof. Denote $M_f = M \otimes^{\mathbf{L}}_R R_f$ and choose a distinguished triangle $F \to M \to M_f$. By assumption $H^i(M_f) = H^i(M)_f = 0$ for $i < 0$. Thus it suffices to show that $H^i(F) = 0$ for $i < 0$. Note that $H^i(F)$ is f-power torsion for all $i \in \mathbf{Z}$. On the other hand, since $f : M_f \to M_f$ is an isomorphism, we see that C is isomorphic to the cone of $f : F \to F$ (use Derived Categories, Proposition 05R0). Now, if $H^i(F) \neq 0$, then the kernel of $f : H^i(F) \to H^i(F)$ is nonzero, which implies that $H^{i-1}(C)$ is

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nonzero. Our assumption implies this cannot happen if $i - 1 < -1$ which finishes the proof. $\square$